

Ashitha Paramesha Gowda

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Summary

Backend-focused full stack engineer with 3+ years of experience building production-grade systems using **Node.js**, **NestJS**, **PostgreSQL**, **AWS** & **Python**. I specialize in **microservices**, **event-driven architectures**, & backend systems that support **real-time**, **AI-integrated workflows**.

Education

Johns Hopkins University, Baltimore, MD

Aug 2023 – May 2025

MS in Computer Science (3.5/4)

Relevant Coursework: Computer Architecture, Applied Cryptography, Security and Privacy in Computing, Future Networks (5G, SDNs), Foundations of AI, Software System Design, Software Engineering.

Vidyavardhaka College of Engineering, Karnataka, India

Aug 2017 – May 2021

Bachelor of Engineering in Information Science (3.8/4)

Experience

Software Engineer - Research Assistant

Mar 2025 – Present

Johns Hopkins University, Baltimore, MD

- Built a **Python-based, layout-driven extraction LLM** with **200+ samples** (70+ fields each); **replaced 15+ pipelines** by designing **reusable schemas** that enabled **field-level accuracy** across unseen Medicaid PDFs.
- **Fine-tuned** Gemini 2.0 on the custom dataset & deployed it via **Vertex AI**; automated full-document parsing across 15+ Medicaid waiver types with **85% accuracy**, boosting structured data coverage from **15 fields to 70+** & saving **100+ analyst hours**.

Software Engineer II

May 2023 – July 2024

JKTech, Bengaluru (Mid-sized - **B2B SaaS** firm, **Project** : Inveniam.io – data platform for private real estate equity)

- Built an event-driven sync system using **Node.js (NestJS)**, **TypeScript**, & **PostgreSQL** to track & sync file meta-data & cloud pointers; implemented **hash-based change detection** & recorded updates on-chain for **blockchain-based file verification**, reducing sync lag by 50%.
- Leveraged **EventStore** to replay events & trace production issues; resolved **80% of support tickets**, including **30% classified as critical**, by enabling version-aware debugging of folder metadata & sync flows.

Software Engineer I

Jan 2022 – Apr 2023

Juego Studios, Bengaluru (Startup - **B2B2C gaming SaaS** firm)

- Architected a **real-time** voice/video communication **plugin for the NJS2** engine using **WebRTC** with **Socket.io**; **supported 1:1 & 1:N chat** with **400+ concurrent users**, & **reduced dev effort by 80%** in Node.js.
- Resolved **critical backend blockers** like race conditions (via **deque logic**) & **pot distribution bugs** in card games, **saving 250+ dev days**, preventing a code rewrite & **accelerating card game delivery** by 70%.

Founder, Full Stack Developer

Jan 2021 – Present

CNotes.in, Bengaluru (Startup - **B2B SaaS** firm)

- Built & scaled a CRM & billing platform (**PHP, MySQL**) serving **30K+ subscribers**, replacing manual workflows with real-time forecasts (0.4–5% error) & driving **95% retention**.

Founder, Full Stack Developer

Mar 2025 – Present

Dear: AI-driven Voice Assistant for Memory Loss (Personal Project, Ongoing)

- Architected a **Dockerized, service-oriented backend (Node.js, FastAPI, ChromaDB, Redis)** powering an AI voice agent with real-time audio transcription, memory parsing, and retrieval-augmented dialogue generation.
- Designed a **memory agent** to parse user inputs, fill context gaps with clarifying questions, and enrich memories via **Gemini-driven sentiment analysis and tagging**, producing structured embeddings for long-term retrieval.
- Built a modular **audio pipeline** with custom S3-like storage, chunked uploads, **Whisper** transcription (**OpenAI STT**), and lazy-loaded waveform playback for low-latency voice interactions on mobile and web.

Skills

Languages & Frameworks: JavaScript, TypeScript, Python, SQL, Java; Node.js, NestJS, Express, Django, Flask, React, Next.js

Databases & Infra: PostgreSQL, MySQL, MongoDB; AWS (Lambda, CloudFormation), GCP, Docker, Kubernetes, Jenkins, GitHub Actions

AI/ML: Gemini, Vertex AI, OpenAI APIs, Hugging Face, AssemblyAI; LLM pipelines, JSONL, Textlocal API

Architecture & Tools: REST, GraphQL, WebSockets, OAuth2, JWT, Kafka, RabbitMQ, Microservices, TDD,DDD, Jira